for neural network machine learning. Initially, it was used by Google only, but since 2018, Google has made it available for third-party use. It supports TensorFlow code and allows you to run your own programs on the TPU on Google Cloud. The documentation provided by Google is comprehensive and includes several user guides. The TPU is specifically designed for vector-by-matrix multiplication, an operation that happens multiple times in any ML application.

On-edge computing and AI

On-edge computing gives an engine the power to compute or process locally with low latency. This leads to faster decision-making due to faster response time. However, the biggest advantage would be the ability to send just the processed data to the cloud. Hence, with edge computing, we will no longer be as dependent on the cloud as we are today. This means that lesser cloud storage space is required, and hence lower energy usage and lower costs.

The power consumed by AI and ML applications is no joke. So, deploying AI or machine learning on edge has disadvantages in terms of performance and energy that may outweigh the benefits. With the help of on-edge AI accelerators, developers can leverage the flexibility of edge computing, mitigate privacy concerns, and deploy their AI applications on edge.

Boards that use AI accelerator

As more and more companies make a mark in the field of AI accelerators, we will slowly witness a seamless integration of AI into IoT on edge. Companies like NVIDIA are, in fact, known for their GPU accelerators. Given below are a few examples of boards that feature AI accelerators or were created specifically for AI applications.

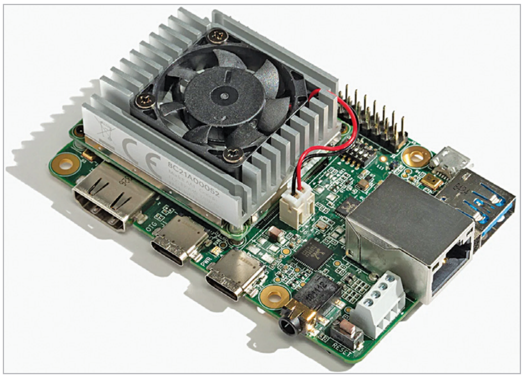

Figure 5: Coral Dev Board (Credit: Coral)

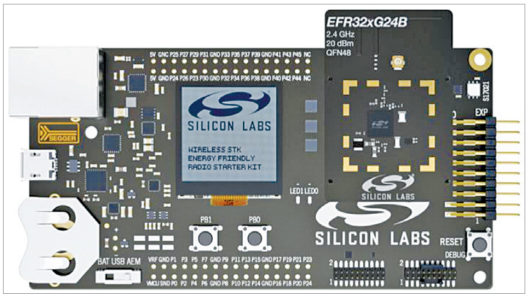

Figure 6: The complete Pro Kit for the new BG24 and MG24 SoCs with all the necessary hardware and software for developing high-volume, scalable 2.4GHz wireless IoT solutions (Credit: Silicon Labs)

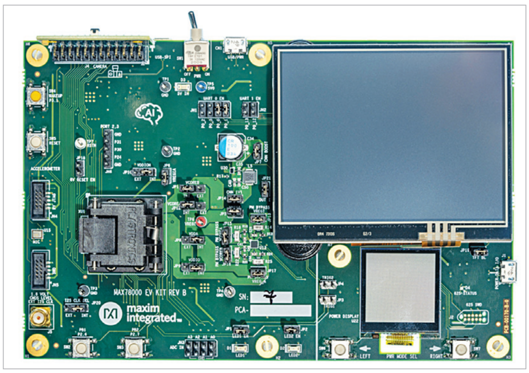

Figure 7: Evaluation Kit for the MAX78000 (Credit: Maxim Integrated)

Google's Coral Dev Board: Google's Coral Dev Board is a single-board computer (SBC) featuring an edge TPU. As mentioned before, TPUs are a type of AI accelerator developed by Google. The edge TPU in the Coral Dev Board is responsible for providing high-performance ML inferencing with a low power cost.

The Coral Dev Board supports TensorFlow Lite and AutoML Vision Edge. It is suitable for prototyping IoT applications that require ML on edge. After successful prototype development, you can even scale it to production level using the onboard Coral system-on-module (SoM) combined with your custom PCB. Coral Dev Board Mini is Coral Dev Board's successor with a smaller form factor and lower price.

BG24/MG24 SoCs from Silicon Labs: In January 2022, Silicon Labs announced the BG24 and MG24 families of 2.4GHz wireless SoCs with built-in AI accelerators and a new software toolkit. This new co-optimised hardware and software platform will help bring AI/ML applications and wireless high performance to battery-powered edge devices. These devices have a dedicated security core called Secure Vault, which makes them suitable for data-sensitive IoT applications.

The accelerator is designed to handle complex calculations quickly and efficiently. And because the ML calculations are happening on the local device rather than in the cloud, network latency is eliminated. This also means that the CPU need not do that kind of processing, which, in turn, saves power. Its software toolkit supports some of the most popular tool suites like TensorFlow.

"The BG24 and MG24 wireless SoCs represent an awesome combination of industry capabilities, including broad wireless multiprotocol support, battery life, machine learning, and security for